



SECP2523
DATABASE (WBL)
SECTION 02

ASSIGNMENT 1

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QUESTION 1

One of the application areas of database systems is e-commerce such as online shopping.

What are the data or information stored in the database system of an online shopping website? Give FOUR (4) of them.

Data	order	Customer
Information	Iphone 16	Edward

QUESTION 2

Answer the questions below based on the following relational schema:

SCIENTIST	(<u>Sc_ID</u> , Sc_Name, Sc_Phone, Sc_Add)	
EXPERIMENT	(<u>Exp_NO</u> , Ex_Name, <u>Proj_Code</u>)	primary key - form by computer - num and alphabet only
PROJECT	(<u>Proj_Code</u> , Proj_Name, Proj_StartDate)	foreign key - shadow key
SCIENTIST_EXPERIMENT	(<u>Sc_ID</u> , <u>Exp_NO</u> , Date, Time)	served as primary key

- a) For each table, identify the primary key and foreign key. If a table does not have a primary key or foreign key, write – in the space provided. (4M)

TABLE	PRIMARY KEY	FOREIGN KEY
SCIENTIST	Sc_ID	—
EXPERIMENT	Exp_NO	Proj_Code
PROJECT	Proj_Code	—
SCIENTIST_EXPERIMENT	—	Sc_ID, Exp_NO

- b) Does the table exhibit entity integrity? Answer 'Yes' or 'No' and then explain your answer. (4M)

TABLE	ENTITY INTEGRITY	EXPLANATION
SCIENTIST	Yes	have primary key
EXPERIMENT	Yes	have primary key
PROJECT	Yes	have primary key

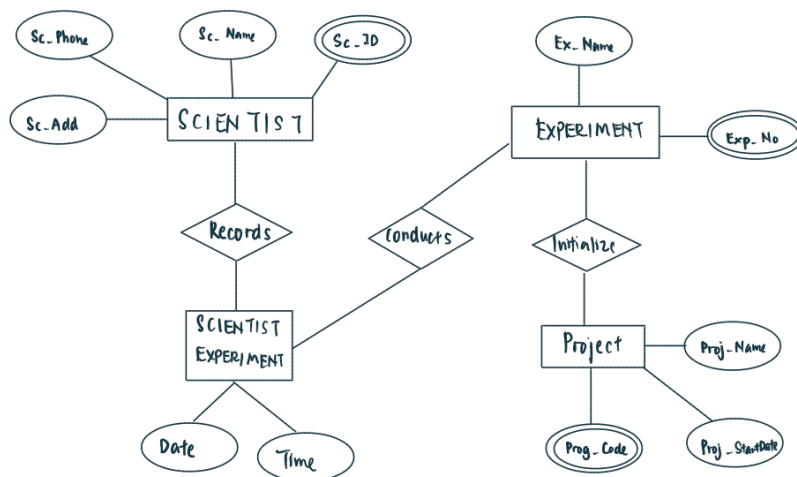
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SCIENTIST_EXPERIMENT	No	don't have a primary key
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- c) Does the table exhibit referential integrity? Answer 'Yes' or 'No' and then explain your answer. (4M)

TABLE	REFERENTIAL INTEGRITY <i>entity have foreign key</i>	EXPLANATION
SCIENTIST	No	Doesn't have foreign key
EXPERIMENT	Yes	Have foreign key
PROJECT	No	Doesn't have foreign key
SCIENTIST_EXPERIMENT	Yes	Have foreign key

- d) Draw ERD using Chen notation based on the given relational schema. (4M)



QUESTION 3

Read the following and complete the task listed.

Consider the following about Luxury Records:

Here is the information that you gather:

- Luxury has decided to store information about musicians who perform on its albums (as well as other company data) in a database.
- Each **musician** that records at Luxury has an SSN, a name, an address, and a phone number.
- Poorly paid musicians often share the same address, and no address has more than one phone.
- Each **instrument** used in songs recorded at Luxury has a unique identification number, a name (e.g guitar, synthesizer, and a flute) and a musical key (e.g., C, B-flat, E-flat).
- Each **album** recorded on the Luxury label has a unique identification number, a title, a copyright date, a format (e.g., CD or MC) and an album identifier.
- Each **song** recorded at Luxury has a title and an author.
- Each musician may play several instruments, and a given instrument may be played by several musicians.
- Each song album has a number of songs on it, but no song may appear on more than one album.
- Each song is performed by one or more musicians, and a musician may perform a number of songs.
- Each album has exactly one musician who acts as its producer.
- A musician may produce several albums, of course.

Task:

1. List each Entity that is identified in Luxury Records System.
2. For each entity list the attributes.
3. Draw a logical model to represent each entity and the relationships between them.
4. Draw an Entity Relationship Diagram (ERD) that captures the entire system.

1) musician , instrument , album and song

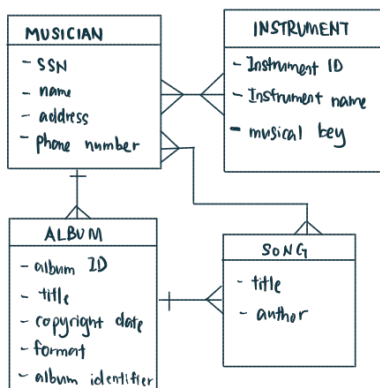
2) MUSICIAN (SSN , name , address , phone number)

INSTRUMENT (Ins-ID , Instrument name , musical key)

ALBUM (album-ID , title , copyright date , format , album identifier)

SONG (title , author)

3 & 4)



QUESTION 4

Give an example of each of the three types of relationships.

- i) One-to-many relationship
One man own many car
- ii) Many-to-many relationship
many customer can buy many products
- iii) One-to-one relationship
Student have one program
- iv) Many-to-one relationship
many car owned by one man

QUESTION 5

CLIENT (Client_Number, Client_Name, Street, City, State, Postal_Code, Amount_Paid, Current_Due, Recruiter_Number)

RECRUITER (Recruiter_Number, Last_Name, First_Name, Street, City, State, Postal_Code, Rate, Commission)

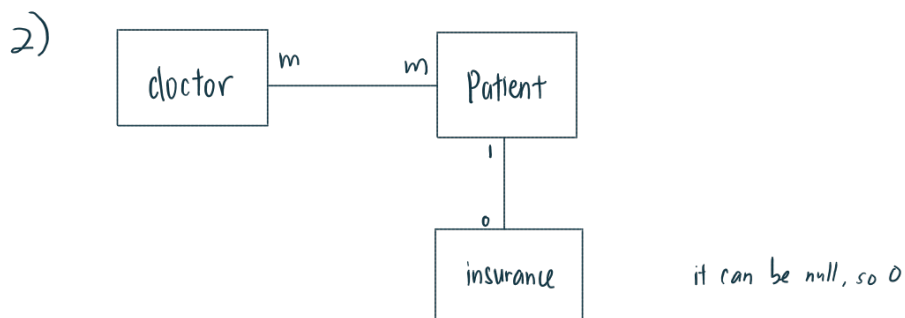
- a) What is the relationship between Recruiter and Client?
many-to-one
- b) What is the primary key for the Client table and Recruiter table?
Client-Number is primary key for Client table
Recruiter-Number is primary key for Recruiter table
- c) Define foreign key? Identify foreign key in the Client table.

Foreign key is an attributes that served as primary key of another table
Recruiter-Number

QUESTION 6

Draw the Chen & Crow's Foot notation based on the following statements:

- i) "A customer can make many payments, but each payment is made by only one customer." *many to one*
- ii) "A **medical doctor** may treat many **patients**. Some patients may be treated by more than one medical doctor. A patient may have or may not have insurance plan."



QUESTION 7

Consider the following relations for a database that keeps track of student enrollment in courses and the books adopted for each course:

STUDENT (SSN, Name, Major, Bdate)

COURSE (Course#, Cname, Dept)

ENROLL (SSN, Course#, Quarter, Grade)

BOOK_ADOPTION (Course#, Quarter, Book_ISBN)

TEXT (Book_ISBN, Book_Title, Publisher, Author)

Draw a relational schema diagram specifying the foreign keys for this schema. ERD

